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An optimized machine learning approach for short-term electricity demand forecasting considering environmental and historical influencing factors

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Abstract
This study presents an optimized machine learning framework for short-term electricity demand forecasting to support energy purchase planning and operational reliability of power distribution systems—particularly under market restructuring and climate variability. The proposed method employed a Random Forest (RF) algorithm enhanced through Grid Search hyperparameter tuning. Temperature data, as a key environmental factor, along with previous-day electricity demand, were used as input features. The model was evaluated using real-world data from a distribution substation in an Iranian city and was compared with conventional forecasting approaches, including Exponential Smoothing (ES), Seasonal ARIMA (SARIMA), and the standard RF. The optimized RF model achieved a Mean Square Error (MSE) of 0.01 and an R^2 of 0.894 in winter, outperforming the other methods across all seasons. These results confirmed that systematic hyperparameter optimization can significantly enhance the predictive performance of machine learning-based models for day-ahead electricity load forecasting.

Keywords Machine learning, Short-term forecasting, Electricity demand, Time-series forecasting, Random forest, Seasonal autoregressive integrated moving average (SARIMA)

1 Introduction

1.1 Motivation

Electricity has been recognized as a fundamental necessity for human livelihood and national development. In recent years, governments, researchers, and industrial organizations have paid increasing attention to electricity production, consumption, and supply due to their significant economic and social implications. Traditionally, electricity was generated using primary energy sources such as fossil fuels, nuclear power, and renewable energy. However, fossil fuels have continued to account for a major portion of global electricity generation, resulting in elevated levels of greenhouse gas (GHG)

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emissions and serious environmental consequences. Addressing these issues has been defined as a core objective of the United Nations Sustainable Development Goals (UNSDGs), which promote reductions in global carbon emissions and the adoption of renewable energy sources [1, 2].

Accurate electricity demand forecasting has emerged as a strategic tool for reducing GHG emissions and optimizing power generation by preventing both overproduction and undersupply. Since electricity consumption varies significantly across regions depending on access levels and socioeconomic development, the Human Development Index (HDI) has been commonly used as an indicator to reflect this relationship [3, 4]. Precise demand estimation enables more effective balancing between electricity supply and demand, ensuring adequate and equitable distribution while minimizing energy waste and supply shortages. Furthermore, it supports strategic energy production planning and enhances operational efficiency [5, 6].

1.2 Forecasting horizons and STLF importance

Electricity load forecasting is typically categorized into three main types based on the prediction time horizon [7–9]:

- *Short-Term Load Forecasting (STLF)*: from a few hours to one week ahead.
- *Medium-Term Load Forecasting (MTLF)*: ranging from several weeks to a few months.
- *Long-Term Load Forecasting (LTLF)*: spanning from several months to years.

Among these, STLF has played a particularly vital role in energy management and power system operation. It has been widely utilized in applications such as real-time dispatching, electricity market planning, load shedding decisions, and short-term energy trading. The accuracy and reliability of STLF models have had a direct impact on both the economic and operational performance of modern power systems.

In summary, the importance of accurate, efficient, and scalable load forecasting systems is underscored by the growing global demand for electricity, environmental concerns over fossil fuel usage, and the strategic goals outlined by the United Nations Sustainable Development Goals (UNSDGs). Among the various forecasting horizons, a pivotal role in operational decision-making and energy efficiency is played by short-term load forecasting (STLF). Therefore, a robust, interpretable, and optimized machine learning model was developed in this study—based on Random Forest and tuned using Grid Search—to enable next-day electricity demand prediction from temperature and historical load data. The proposed model was designed to enhance the accuracy and practicality of STLF in real-world power distribution systems, thereby ensuring more sustainable and intelligent energy management.

1.3 Literature review

We can divide the methods presented for predicting electric charge into two categories: classic and artificial intelligence-based methods. Some examples of classical methods used in load forecasting include exponential smoothing (ES) [10], autoregressive integrated moving average (ARIMA) [11], autoregressive moving average (ARMA) [12], and Kalman filter (KF) [13]. The implementation and application of these methods are simple and do not require a lot of computational load; But the output of these methods

usually does not have the necessary accuracy. Univariate models based on patterns of daily cycles of load time series and linear regression are proposed in [14] for short-term load forecasting. By removing trends and seasonal fluctuations from periods longer than daily, the patterns utilized as input and output variables make the forecasting problem easier to understand.

In recent years, a variety of methods—both classical, such as Generalized Additive Models (GAMs), and modern techniques, including artificial neural networks, deep learning [15–17], fuzzy logic [18, 19], and expert systems [20]—have been applied to predict electricity demand, each offering different strengths depending on the context and data characteristics. Despite the increase in accuracy compared to classical methods, these methods also have their disadvantages. For example, choosing the learning method, proper setting of the number of layers or the number of neural network neurons, optimal settings of membership functions and rules in fuzzy logic are among the challenges of these methods. Also, by increasing auxiliary inputs to the problem such as temperature, sunlight and holidays, the settings of those methods become more complicated. Anyway, choosing the right method according to the advantages and disadvantages of each of them still needs more study. A survey of various deep learning models for short-term load forecasting was conducted by the authors in [7]. They used three sets of actual data to assess seven deep learning models. They demonstrated that there is no clear relationship between key findings and model complexity. In certain instances, short-term load forecasting can yield superior outcomes with more basic models. To increase the accuracy of forecasting, a new deep neural network framework that combines the hidden feature of the convolutional neural network (CNN) model with the LSTM model is proposed in [21]. To verify the applicability and stability of the suggested concept, extensive tests were carried out in real-world scenarios. The CNN and LSTM models were contrasted with the suggested model's forecasting abilities. A parallel CNN- recurrent neural network (RNN) model was suggested by the author in [22] in order to forecast load consumption one day in advance. Regression-based models, CNN-RNN and DNN, performed worse than temperature, days of the week, hours of the day, and holidays when utilized as features for the historical load series. A precise deep neural network technique for STLF is presented in [23]. Five artificial intelligence algorithms that are frequently used in load forecasting are compared to see how well the suggested algorithm forecasts. Using Root Mean Square Error (RMSE) and Mean Absolute Percentage Error (MAPE), they contrasted the output of the suggested model with five different machine learning methods. For short-term load forecasting (STLF), a mixed approach based on convolutional neural networks (CNN) and fuzzy time series (FTS) is suggested [24]. As a result, the suggested method transformed multivariate time series data—such as hourly load, hourly temperature, and fuzzified load time series—into multi-channel images that were then fed into a suggested deep learning CNN model with the appropriate architecture.

Recently, the use of machine learning methods for prediction problems has attracted the attention of researchers [25]. Machine learning based on decision trees such as random forest (RF), support vector machine (SVM), and regression-based machine learning such as multivariate adaptive regression spline (MARS) are among the machine learning methods that have been used due to their advantages over other methods. RF model is a univariate model composed of regression trees and has few parameters to adjust [26].

SVM is a very effective and efficient model in machine learning that has the ability to solve non-linear problems despite the small learning data and is a good option for forecasting applications. MARS is a new machine learning tool that has speed and flexibility suitable for forecasting applications. In an Internet of Things (IOT) context [27], offers a novel online load forecasting method utilizing supervised machine learning (ML) methods. Through the use of ML algorithms, this work offers the possibility of STLF online with precise prediction models. In a research lab at JNTUH, Hyderabad, weather and electricity load consumption data are used to train machine learning algorithms in order to implement STLF [28]. offers a general approach based on support vector regression machines (SVR) for short-term load forecasting (STLF). Two significant enhancements to the SVR-based load forecasting technique are presented: the process for creating model inputs and the subsequent feature selection algorithms-based model input selection. The suggested approach includes minimizing operator contact during the model-building process as one of its goals. To address the STLF issue, a support vector machine (SVM) model is put out [29]. In [29], a new learning technique, is effectively applied to load forecasting. In addition, additional tests and in-depth assessments are carried out and provided here, inspired by the competition outcomes and other competitors' strategies. A new hybrid yearly power load forecasting model based on the Moth-Flame Optimization algorithm (MFO) and least squares support vector machine (LSSVM) is proposed in [30]. The MFO method is a cutting-edge nature-inspired metaheuristic algorithm that optimally determines the parameters of the LSSVM model. In the meantime, this investigation also makes use of the rolling mechanism. A unique LSSVM for the efficient prediction of daily building energy consumption is proposed in [31] by utilizing the DSORCGA, a mix of the real-coded genetic algorithm (RCGA) and the direct search optimization (DSO) method. The suggested DSORCGA is used to optimize the free parameters of LSSVM for quicker computation times and better prediction accuracy. It differs from the traditional RCGA in terms of the reproduction operator and the crossover operator.

1.4 Contributions

Although significant progress had been made in applying artificial intelligence (AI) and machine learning (ML) techniques to short-term load forecasting (STLF), several challenges remained unresolved, including the trade-off between model complexity, interpretability, and computational efficiency. Furthermore, there was a lack of studies focusing on the systematic optimization of ensemble learning models such as Random Forest (RF) through hyperparameter tuning, which was considered critical for enhancing forecasting performance.

To address these gaps, this study proposed a novel STLF model based on an optimized Random Forest algorithm. Unlike traditional approaches, the proposed method integrated temperature and historical load data to effectively capture both seasonal and daily load variations. Grid Search optimization was applied to tune key hyperparameters—including the number of trees, tree depth, and feature subset size—resulting in a model that achieved a superior balance between predictive accuracy, robustness, and computational efficiency.

When compared to classical statistical models and unoptimized RF implementations, the optimized model demonstrated substantial improvements in forecasting accuracy

and stability, as validated by experiments on real-world datasets. Moreover, in contrast to more complex deep learning frameworks, this approach offered greater interpretability and reduced computational burden without compromising accuracy, making it highly practical for real-time forecasting applications in power distribution systems. Unlike many existing works that evaluate forecasting models on limited or single-season datasets, this study conducts a seasonal performance assessment across three different days in summer, autumn, and winter. This approach provides a clearer understanding of the model's stability, adaptability, and generalization capability under diverse environmental conditions.

The main contributions of this research are as follows:

1. *Application of the Random Forest algorithm for STLF*: Capable of handling multidimensional and nonlinear relationships between temperature and load.
2. *Model optimization using Grid Search*: Hyperparameter tuning to systematically improve prediction accuracy.
3. *Seasonal performance analysis of the proposed model for STLF*: Allows assessment of the model's reliability across different environmental conditions by testing it on representative days from three distinct seasons (summer, autumn, and winter), highlighting its robustness and generalization capability.

This article is presented as follows: The methodology is fully presented in Sect. 2. Sections 3 and 4 presents the process of training and testing and simulation results, and Sect. 5 of the article deals with conclusions and ends.

2 Proposed methodology and evaluation criterion for demand forecasting

2.1 Methodology

In this section, we introduce 3 algorithms used to predict the next day's demand.

- *Random Forest*

Random Forest is an ensemble learning method that is used for classification and regression and uses the combination of several decision trees. This algorithm was introduced by Bremen in 2001 and is based on bootstrap techniques and random feature selection [32, 33]. The main goal of this algorithm is to reduce the variance of the model and improve the accuracy of predictions by combining the predictions of weak models (weak learners) in parallel.

Since the forecasted variable directly depends on the scores of each decision tree, as mentioned in (1), the difficulty that comes with having numerous decision trees contributes to the RF identifies it as a black box technique.

$$\widehat{y} = \sum_{l=1}^{t} \frac{p\left(\widehat{y_t}|x\right)}{t} \tag{1}$$

where t is the number of trees, $\widehat{\mathrm{y}}$ is the forecasted variable, and $\mathrm{p}\left(\widehat{\mathrm{y}}_{\mathrm{t}}|\mathrm{x}\right)$ is the forecasted variable's probability function given the independent factors. As a result of the decision trees' dependence, the objective function J takes the complexity of the set of decision trees into account in addition to attempting to minimise the error, as demonstrated in (2).

$$j = \sum_{k=1}^{n} \frac{(\widehat{y_k} - y_k)^2}{n} + \sum_{l=1}^{t} f(\widehat{y_k}) \tag{2}$$

where $\widehat{y_k}$ is the predicted variable at time k, n is the number of samples, and the function $f(\widehat{y_k})$ determines the difficulty of each decision tree. The advantages of the proposed algorithm are as follows.

Variance reduction: By combining multiple decision trees, random forest has less tendency to overfit.

Robust against noise and unbalanced data: This algorithm copes well with noisy data and uncorrelated features.

Importance of features: Random Forest can measure the importance of each feature by reducing criteria such as average accuracy or average Gini. This feature allows researchers to identify more important features.

Ability to evaluate performance with out-of-bag data: Data not used in each tree for training (out-of-bag data) can be used to evaluate model performance without the need for separate test data.

The step-by-step flowchart of random forest algorithm based on Fig. 1 are as follows:

Start: The starting point of the process that represents the beginning of the algorithm execution.

Data (Dataset): The input of the algorithm consists of the original data set to be analyzed and predicted. *Bootstrap Sampling*: In this step, several random subsets are constructed from the original data. For this purpose, sampling method with replacement is used, so that some samples may be selected several times and others may not be selected at all.

Random feature selection: At each node of the decision tree, a random subset of features is selected. This gives more diversity to the trees and avoids excessive correlation between trees.

Growing decision trees: For each subset of data, a decision tree is constructed. These trees are expanded as far as possible to split the data.

Creating multiple decision trees: This step represents the construction of a number of independent decision trees obtained from different subsets and random features.

Aggregating predictions: For the final prediction, the results of all trees are aggregated. In classification problems, majority voting is used, and in regression problems, average predictions are calculated.

Final prediction: The final output of the algorithm is the result predicted by the random forest.

End: The end point of the process that indicates the termination of the algorithm execution.

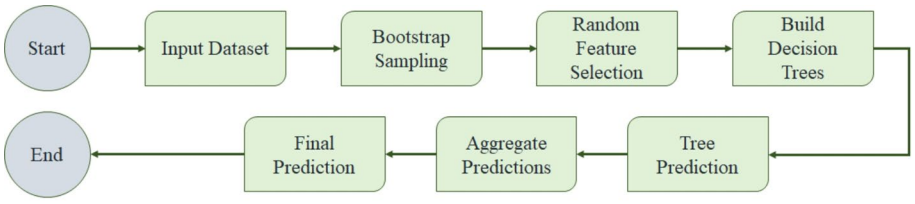


Fig. 1 Flowchart of the proposed method

To enhance understanding of the suggested approach, the optimized RF pseudocode is displayed as follows: Algorithm 1:

Algorithm 1 The pseudo code of the optimized RF algorithm

```
Input:
D: Training data including features and labels
N: Number of decision trees in the forest
M: Number of randomly selected features for each tree (M <= total number of features)
B: Number of random samples from the training data for each tree (Bootstrap samples)
Output:
Random Forest model
Initialize an empty Random Forest: Forest = []
for i = 1 to N:
Randomly select B samples with replacement from the training data (Bootstrap)
Randomly select M features from the total available features
Build a decision tree (Tree) using the selected B samples and M features
Add the tree to the forest: Forest.append(Tree)
For an input sample X:
For each tree in Forest:
Get the tree's prediction: Tree.predict(X)
Collect all the predictions (Vote collection)
Majority voting: Choose the most frequent prediction among the trees as the final result
Return the final prediction as the output of the Random Forest model
```

- *SARIMA*

Seasonal aautoregressive integrated moving average (SARIMA) algorithm is a prediction model for time series that has the ability to analyze seasonal data [34]. This model is divided into three main parts: autocorrelation (AR), variance (I) and moving average (MA).

The general formula of SARIMA is given by Eq. (3):

$$SARIMA_s\,(p,\ d,\ q)\,(P,\ D,\ Q) \tag{3}$$

where, p, d, and q represent the number of autocorrelated lags, number of differences needed to make the data stationary, and number of moving average lags. P, D, and Q represent the number of seasonal autocorrelated lags, number of seasonal differences, and number of seasonal moving average lags, respectively. s is the length of seasonal period. By analyzing past patterns, this algorithm produces predictions for future values and is particularly effective on seasonal data.

- *Exponential smoothing*

Exponential smoothing is a simple and effective method for predicting time series based on historical data [35]. The algorithm uses an exponential weighting for past data, so that more recent data has a greater influence on the prediction. The exponential smoothing model is given by Eq. (4):

$$S_t = \alpha\, x_t + (1 - \alpha\,)_{S_{t-1}} \tag{4}$$

where, $\mathrm{S_t}$, α, $\mathrm{x_t}$, and $\mathrm{S_{t-1}}$ represent the prediction at the time of t, smoothing coefficient (between 0 and 1), Actual value at the time of t, and prediction at the time of t-1. amount of α determines how much weight to give to new data; Values higher than α have more emphasis on newer information.

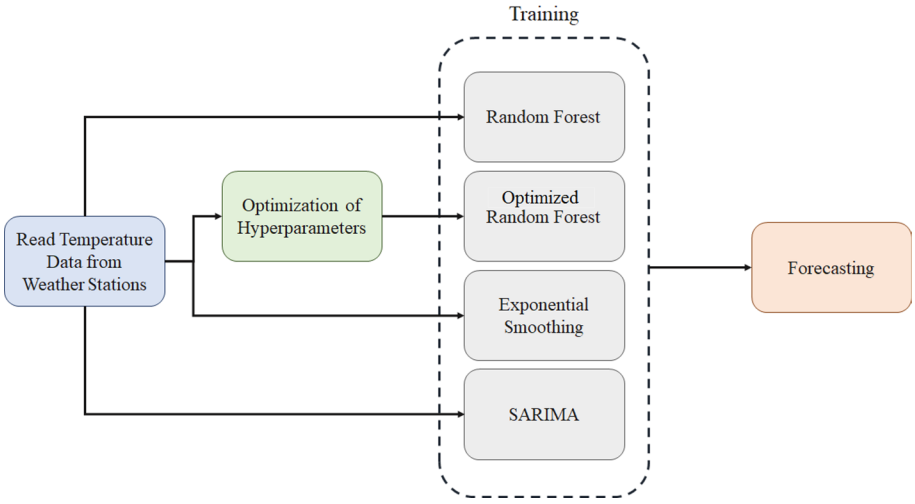


Fig. 2 General prediction block diagram (training and testing) for the desired methods

After introducing and explaining the algorithms mentioned in this study, according to Fig. 2, the general block diagram of the training and testing of four machine learning and classical methods, called RF, optimized RF, SARIMA, and Exponential smoothing using temperature and demand data are shown.

According to Fig. 2, it is clear that temperature and demand data have been used for training and testing different methods. In Sect. 3, the training and testing model of the proposed method is presented and explained in the form of two separate flowcharts. Also, according to this figure, it is clear that the parameters of the random forest optimization algorithm have been optimized by grid search. The parameters of the RF algorithm are as follows:

The number of trees (n_estimators): the more the number of trees, the more accurate the model will be, but the computational cost will also increase. *Depth of trees (max_depth)*: Limiting the depth of trees can prevent overfitting. *Number of selected features (max_features)*: This parameter determines how many features are randomly considered for division in each node. Correct selection of this value helps to improve the performance of the model. The optimized values of the grid search for the *max_depth*, *n_estimators* and *max_features* parameters based on the right and wrong method with the aim of achieving the desired answers in the training phase, are equal to 5, 300 and 9, respectively.

2.2 Evaluation criterion for demand forecasting

In this study, mean square error (MSE) and coefficient of determination criteria (R^2) [5, [6] as defined in Eqs. (5, 6) are used to evaluate the performance of the proposed model.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(\widehat{y_i} - y_i)^2 \tag{5}$$

$$R^2 = \frac{\sum_{i=1}^{n}(\widehat{y_i} - y_{Mean})^2}{\sum_{i=1}^{n}(y_i - y_{Mean})^2} \tag{6}$$

where, n, y_i, and $\widehat{y}_i$ represents the number of data or samples, actual value (original data), and value predicted by the model, respectively [5, 6]. $\mathrm{y}_{\mathrm{Mean}}$ represents the average

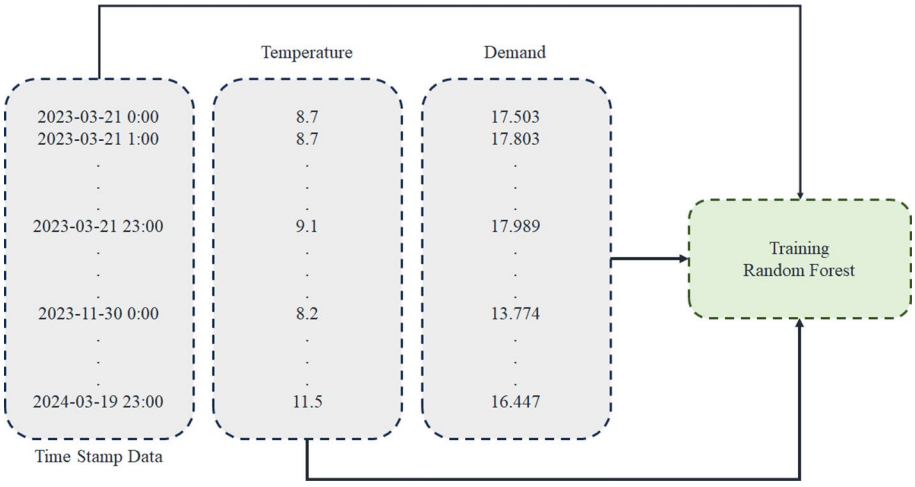


Fig. 3 Training process of the optimized RF

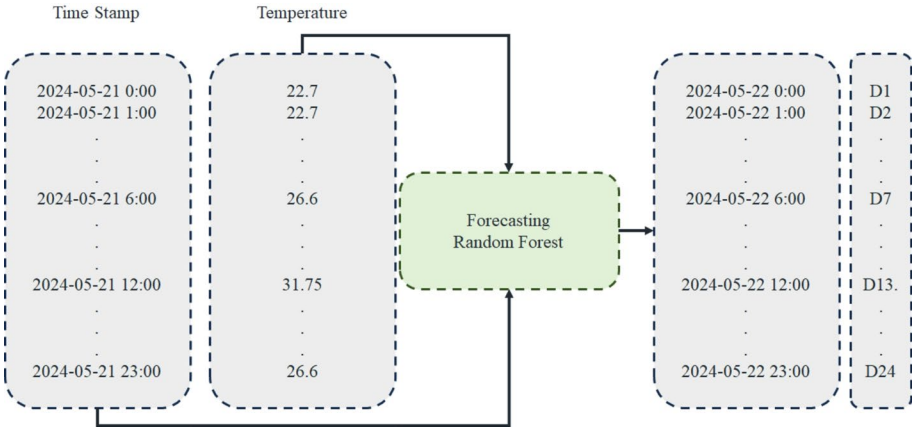


Fig. 4 Testing process of the optimized RF

demand value, while MSE quantifies the discrepancy between the actual and predicted values, as discussed in [5, 6]. The lower the MSE value, the more accurate the model is in forecasting. Moreover, Higher R^2 values indicate a more accurate forecast.

3 The process of training and testing the proposed method

As mentioned in the previous sections, temperature and demand data are used to train and test the desired methods in this study. We have used the meteorological data and data of the electricity distribution substation (including 5 feeders) one of the cities of Iran in a period of one year (2023–2024).

According to Figs. 3 and 4, we have used electricity demand and temperature data in the period of 2023 to 2024 to train the proposed method. We have used the demand and temperature data of the previous day (May 21) to predict the twenty-four-hour demand on May 22. In order to validate the proposed model, 85% of the data have been used for training and the rest for testing the proposed methods in this study. In all methods, in order to compare the results and performance of the methods, the same procedure has been used for training and testing. In order to evaluate the prediction results for all 4 methods, the mean square error criterion has been used.

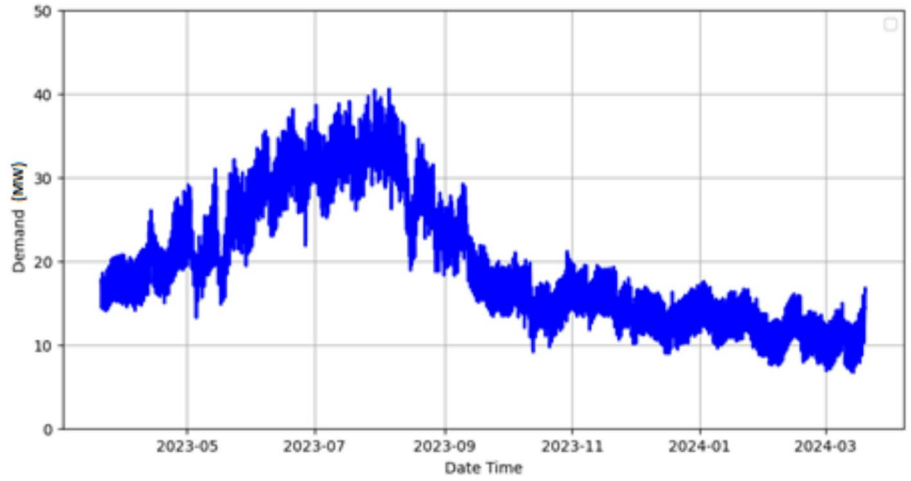


Fig. 5 Hourly time series diagram of electricity demand

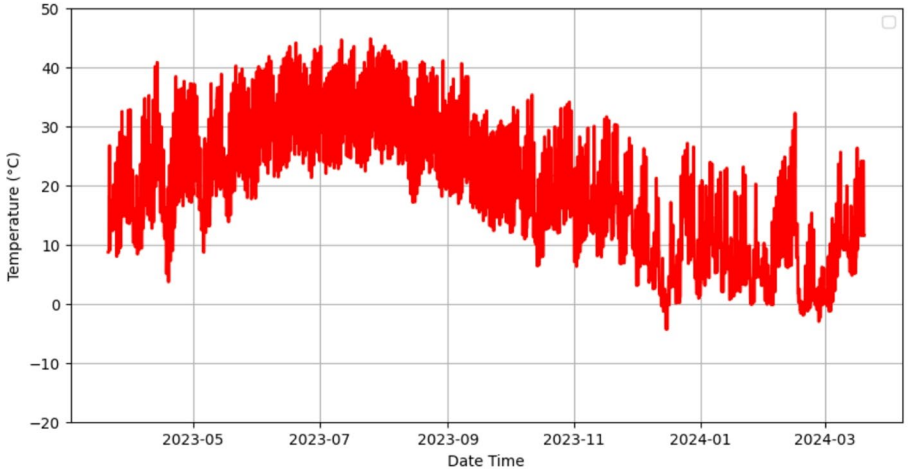


Fig. 6 Hourly time series diagram of temperature

The diagram of the consumed electricity demand (according to data of the electricity distribution substation) and the measured temperature of the weather station in the period of 2023 to 2024 are shown in Figs. 5 and 6.

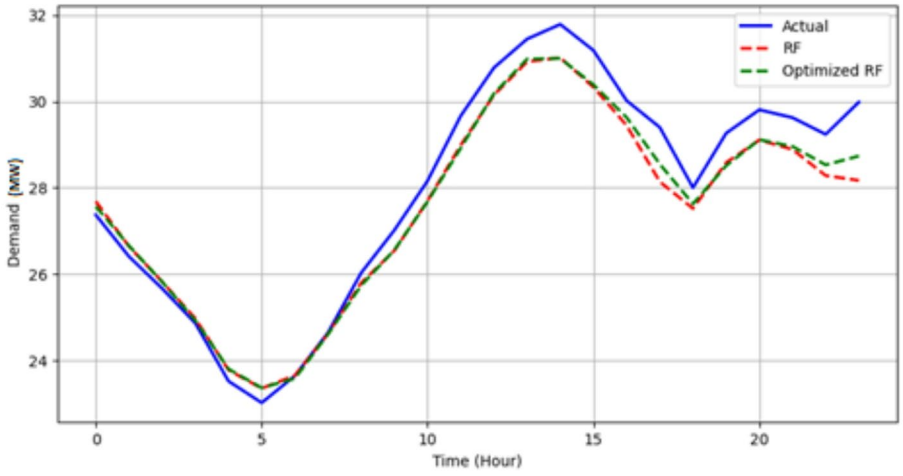
The training and testing process for the optimized RF are shown in Figs. 3 and 4, respectively.

4 Simulation results

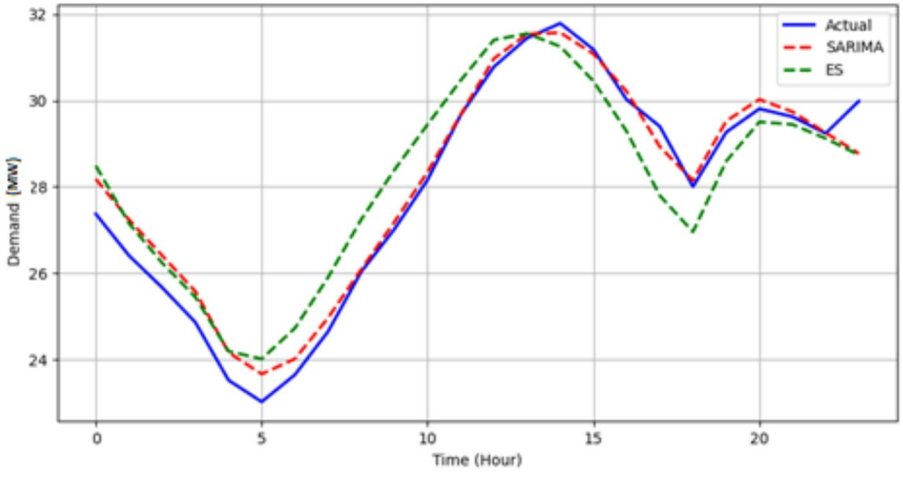
In order to forecast the twenty-four-hour electricity demand, we have chosen one day in each season and have made the forecast for 4 different methods such as, RF, optimized RF, SARIMA, and, Exponential smoothing. All the codes related to teaching and testing different methods in this study are written in Python software.

4.1 Short-term forecasting results

The outcomes of the demand forecast for the following day using the techniques of RF and optimized RF, SARIMA, and ES for the spring and summer seasons, respectively, are displayed in Figs. 7 and 8. Additionally, the results of the demand forecasting for the



(a)



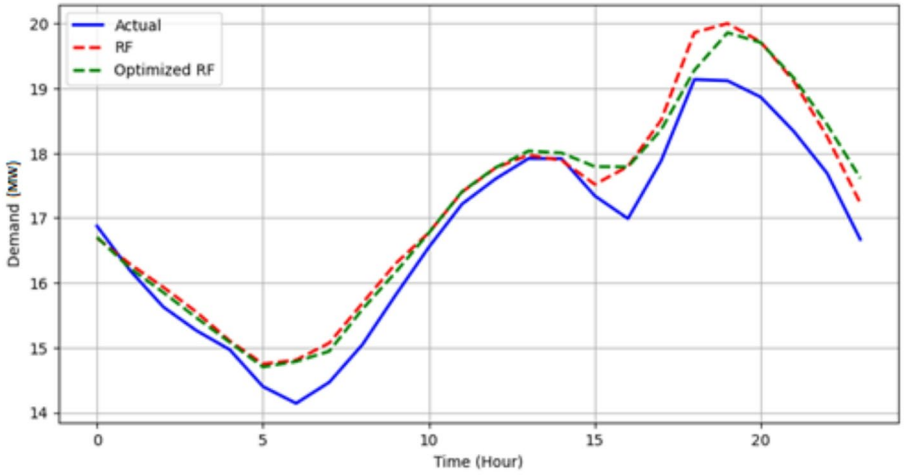
(b)

Fig. 7 Comparison of the next day demand forecasting results from different methods in the spring season (10-06-2024)

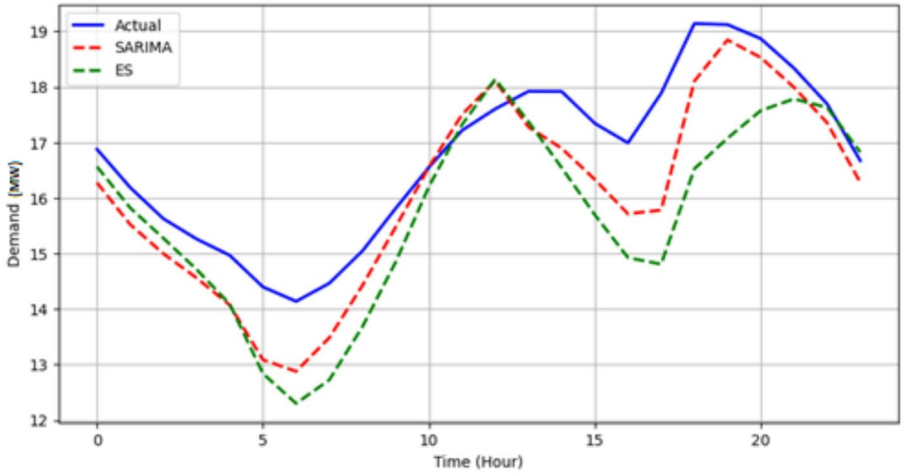
autumn and winter seasons, respectively, obtained from methods RF and optimized RF, SARIMA, and ES, are displayed in Figs. 9 and 10.

Examining the results of short-term demand forecasting (Figs. 7, 8, 9 and 10) shows that the accuracy of the proposed optimized RF is higher than other methods such as RF, SARIMA, and ES. For example, in Fig. 9a, the amount of prediction error of two methods optimized RF and RF in autumn season is very low and the prediction made by these two methods is almost consistent with the actual value of electricity demand.

Examining the results of short-term demand forecasting (Figs. 7, 8, 9 and 10) regarding classical methods shows that SARIMA method performed better than method ES, and the prediction error of this method is less than method ES. In Fig. 7b, we can see the best performance of the SARIMA method in predicting the next day's electricity demand in the autumn season compared to other seasons of the year. Moreover, according to Fig. 10b, the best performance of the ES method in short-term forecasting is related to the winter season compared to other seasons of the year.



(a)



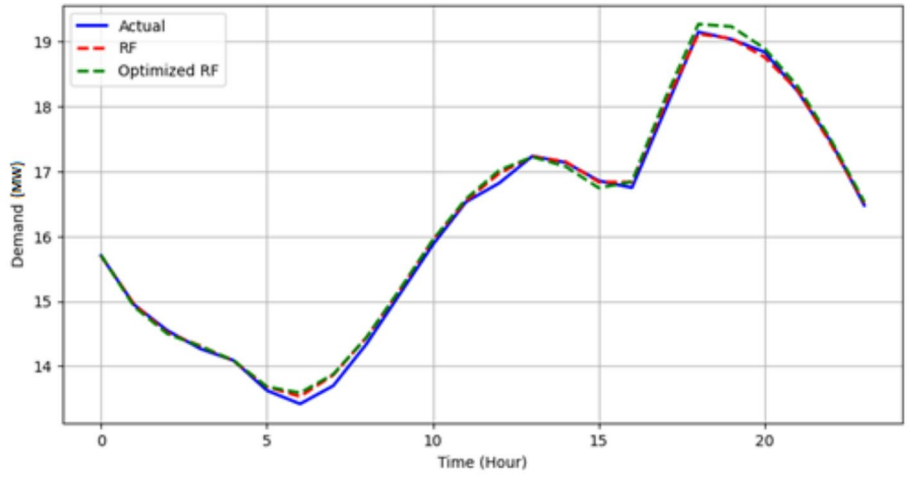
(b)

Fig. 8 Comparison of the next day demand forecasting results from different methods in the summer season (21-09-2024)

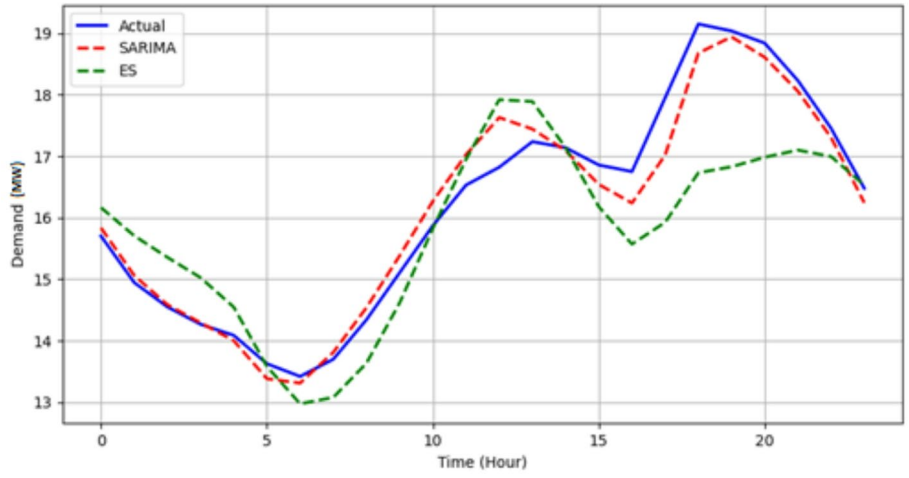
4.2 Evaluation of short-term forecasting results

In this section, the performance of different forecasting methods—including Random Forest (RF), optimized Random Forest (optimized RF), Exponential Smoothing (ES), and Seasonal ARIMA (SARIMA)—is evaluated using standard error metrics across four representative days from different seasons of the year. The evaluation criteria used are the Mean Squared Error (MSE) and the coefficient of determination (R^2), which quantitatively assess the accuracy and goodness-of-fit of each forecasting model.

Tables 1 and 2 summarize the MSE and R^2 values, respectively, for each method on the selected days. As observed, the optimized RF method generally achieves the lowest MSE values, indicating more precise load predictions. Notably, on June 10, 2024, the SARIMA method shows a slightly better MSE; however, this is an exception rather than the norm. For the other three days (September 21, October 28, and January 22), the optimized RF consistently outperforms the other models, confirming its robustness and superior predictive capabilities.



(a)



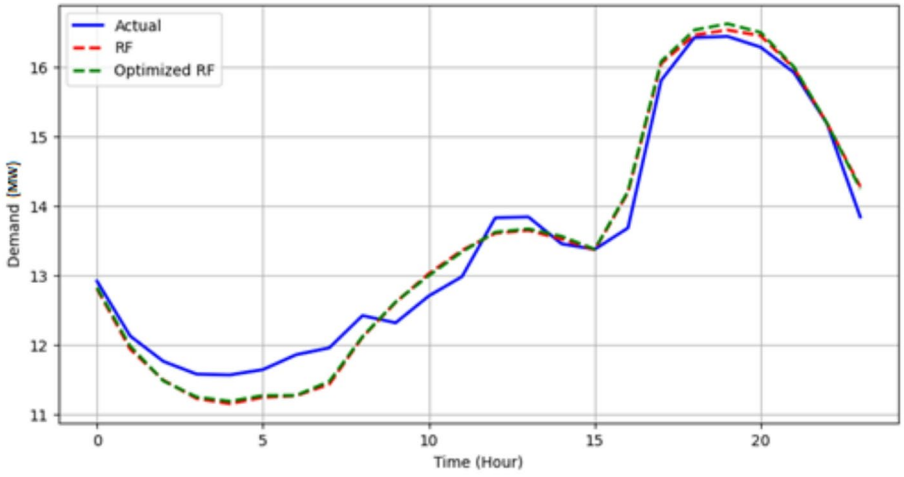
(b)

Fig. 9 Comparison of the next day demand forecasting results from different methods in the autumn season (28-10-2024)

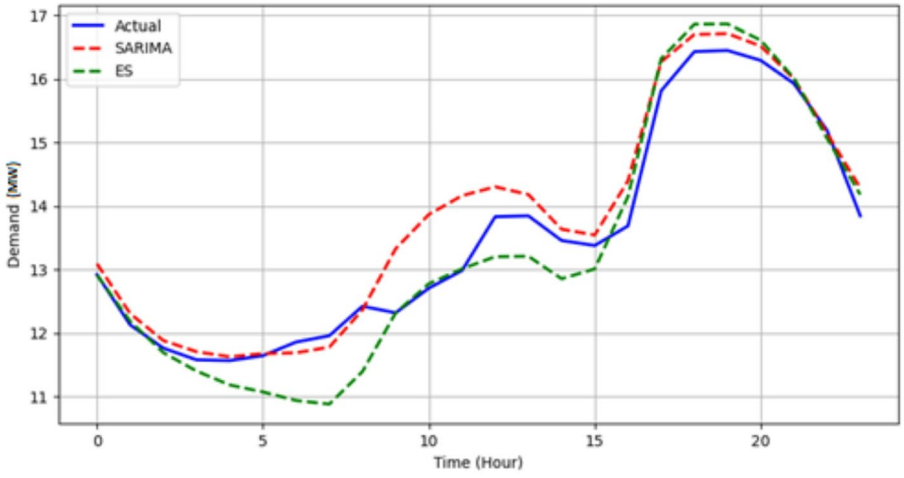
The higher R^2 values for the optimized RF model across all days further emphasize its effectiveness in capturing the underlying nonlinear and complex relationships between temperature and electricity demand. The R^2 values close to 1 indicate a strong correlation between the predicted and actual demand values, reinforcing the reliability of the optimized RF method.

Figures 11 and 12 illustrate the actual versus predicted demand values for June 10, 2024, and January 22, 2024, respectively, using the optimized RF model. These visualizations demonstrate the model's ability to closely track real demand fluctuations, highlighting its practical applicability for short-term load forecasting.

The superior performance of the optimized RF model can be attributed to the hyperparameter tuning via Grid Search, which systematically identifies the best combination of tree numbers, depths, and feature subsets. This optimization reduces overfitting,



(a)



(b)

Fig. 10 Comparison of the next day demand forecasting results from different methods in the winter season (22-01-2024)

Table 1 Comparison of MSE values obtained from different methods in forecasting the next day's demand

Day	MSE (MW)			
	Optimized RF	RF	SARIMA	ES
(10-06-2024)	0.27	0.46	0.22	0.82
(21-09-2024)	0.24	0.26	0.7	1.8
(28-10-2024)	0.01	0.04	0.12	1.12
(22-01-2024)	0.01	0.09	0.22	0.24

improves generalization, and enhances the model's adaptability across various seasonal patterns and weather conditions.

Overall, the results confirm that leveraging historical load and temperature data in an optimized Random Forest framework leads to significant improvements in forecasting

Table 2 Comparison of R^2 values obtained from different methods in forecasting the next day's demand

Day	R^2			
	Optimized RF	**RF**	**ES**	**SARIMA**
(10-06-2024)	0.963	0.915	0.965	0.758
(21-09-2024)	0.998	0.995	0.759	0.751
(28-10-2024)	0.889	0.885	0.745	0.739
(22-01-2024)	0.894	0.887	0.819	0.815

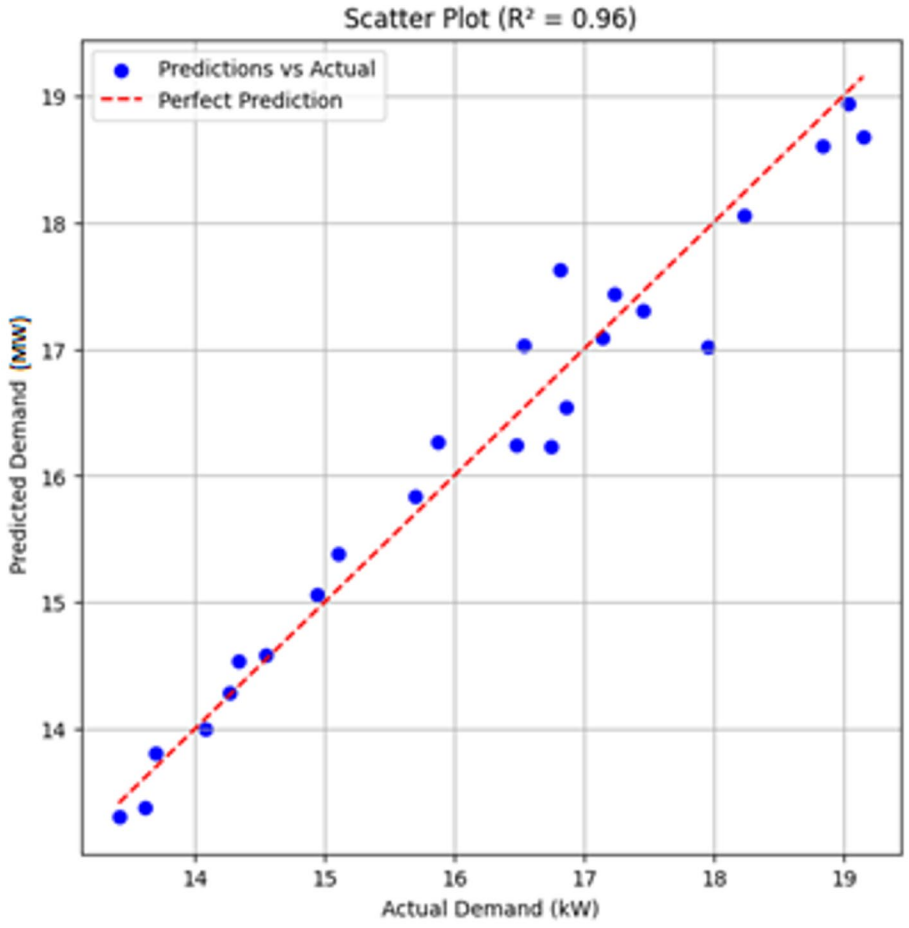


Fig. 11 The distribution of actual and predicted values from the optimized RF for (10-06-2024)

accuracy compared to classical statistical methods and non-optimized machine learning models. This advancement has important practical implications for power grid operators, enabling more efficient load management and energy resource planning.

5 Conclusion

In this paper, a machine learning-based electricity demand forecasting model was developed using temperature and the past 24 hours' demand data as key input variables. The primary objective was to enhance forecasting accuracy within electricity distribution networks, a crucial factor for achieving optimal energy management and minimizing unnecessary resource wastage. The empirical results demonstrated that integrating historical demand data alongside temperature information significantly improves the

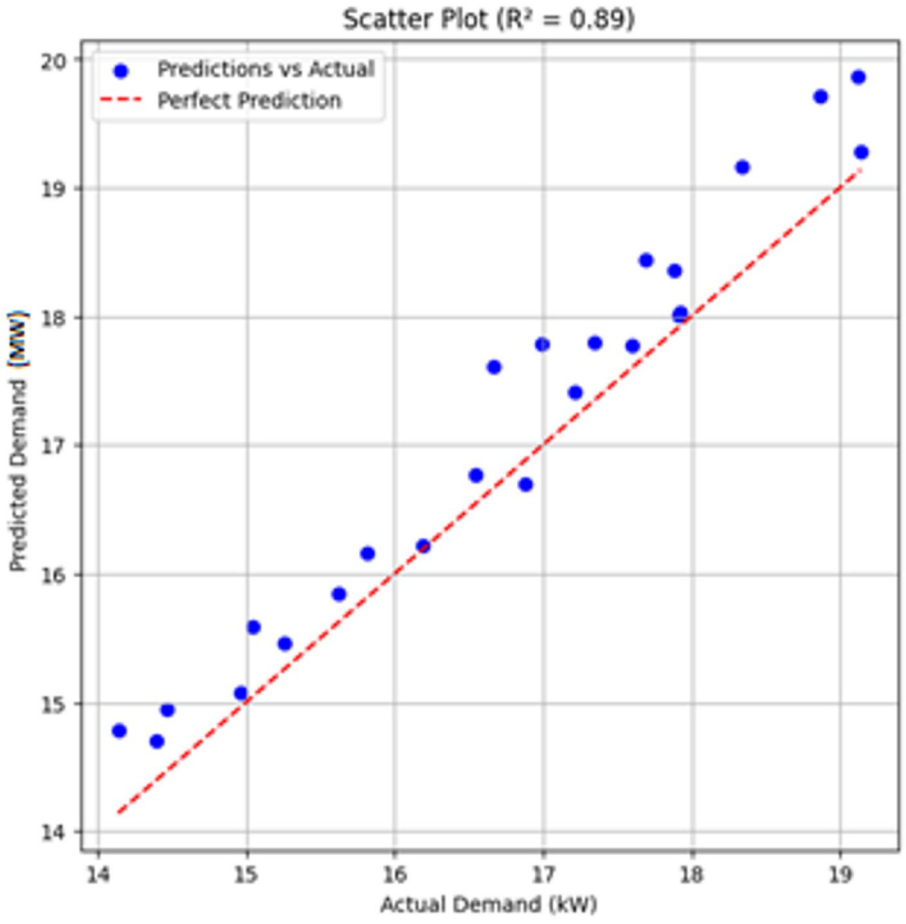


Fig. 12 The distribution of actual and predicted values from the optimized RF for (22-01-2024)

precision of short-term load predictions. Among the various models evaluated, machine learning techniques—especially the optimized Random Forest algorithm—consistently outperformed classical statistical methods by effectively capturing the nonlinear and complex interactions between temperature fluctuations and electricity demand patterns. These findings hold substantial practical implications for power system operators and planners: improved forecast accuracy facilitates more efficient resource allocation, load balancing, and demand-side management, which in turn reduces operational costs and enhances overall grid reliability and stability. Moreover, the inherent adaptability of machine learning models to dynamic environmental changes ensures sustained forecasting performance across a wide range of conditions, making them suitable for real-world deployment. This adaptability also supports faster system responses to unexpected events or seasonal variations.

Practical Implications:

The improved forecasting accuracy provided by the optimized Random Forest (RF) model enables power system operators to anticipate short-term demand fluctuations with greater confidence and precision. This enhanced predictive capability facilitates more efficient resource scheduling, optimal generator dispatch, and better alignment between supply and demand, ultimately leading to reduced operational costs and improved grid reliability. In highly dynamic systems, where load patterns can vary

sharply due to weather, behavioral, or market-driven factors, such predictive accuracy becomes crucial for maintaining system stability and avoiding costly imbalances or emergency interventions.

Limitation and Future Work:

Despite the promising results of this study, some limitations remain. For instance, the current model relies only on historical temperature and past demand data, which may not fully capture other influential factors such as humidity, holidays, or industrial consumption patterns. Moreover, the temperature data used in this study are actual historical values rather than forecasted weather inputs. While this approach is suitable for evaluating model performance under controlled conditions, it does not fully reflect the real-time operational environment of power systems. In practical real-time applications of short-term load forecasting, relying solely on actual (historical) temperature values is unrealistic, as these values are not available ahead of time. Therefore, an important future direction involves incorporating forecasted temperature data as input to the prediction models. Using weather forecasts rather than recorded temperatures will significantly increase the model's real-world applicability and reliability. Although forecasted weather data may introduce a degree of uncertainty, integrating them into the load forecasting framework simulates realistic operational conditions, allowing utility companies to make proactive and data-driven decisions.

For future research, several directions are recommended to enhance the forecasting model's effectiveness and applicability. First, the impact of forecast accuracy on load prediction performance should be systematically assessed, and hybrid methods should be explored to mitigate potential forecast errors. The dataset can be expanded by incorporating more diverse and granular input features, including data from both industrial and residential consumers, to improve the model's accuracy and generalizability. Additionally, hybrid modeling frameworks that integrate the strengths of multiple algorithms could be developed to enhance both prediction precision and robustness. Implementing real-time adaptive learning mechanisms is also encouraged, as they would enable the model to dynamically respond to rapid environmental and behavioral changes, thereby improving its responsiveness under dynamic operating conditions. Finally, the proposed model should be validated across various geographical regions with distinct climatic and consumption profiles to evaluate its scalability and regional adaptability.

Author contributions

H. L.: (1) substantial contributions to the conception or design of the paper; or the acquisition, analysis, or interpretation of data for the paper; P. V.: (1) revising the paper critically for important intellectual content; (2) final approval for publishing the paper; H. R.: (1) the important role for drafting the paper. H. P.: (1) the important role for drafting the paper.

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Data availability

Data will be made available on request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

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Competing interests

The authors declare no competing interests.

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